

Adrian Leong

EDGE AI ENGINEER · EMBEDDED SYSTEMS & MLOPS

Subang Jaya, Selangor, Malaysia

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Education

Monash University

MASTER OF DATA SCIENCE

Sunway, Malaysia

Expected July 2026

Monash University

BACHELOR OF ELECTRICAL AND COMPUTER SYSTEMS ENGINEERING (HONOURS)

Clayton, Australia

2016 - 2019

Technical Skills

Architecture	Computer Vision (OCR/Detection), Geometric Array Filtering, Multi-Model Transformer Orchestration (MiniLM / URL-BERT), Retrieval-Augmented Classification (RAC), XGBoost, Probability Calibration, Hybrid Search (HNSW + Trigram), Reciprocal Rank Fusion (RRF), Explainable AI (XAI), Leave-One-Out (LOO) Matrix Ablation, C++ SHAP Extraction
Edge & HW	ONNX Runtime (Intra-Op Thread Tuning), INT8 Dynamic Transformer Quantization (MiniLM/URLBERT), ARM64 Optimization (Oracle Ampere A1), Embedded C/C++, Linux CFS Context-Switch Optimization, Vector Space Normalization
Deployment	Python, FastAPI, Docker (IPC Tensor Sharing / Shared Memory Tuning), Stateless Inference, Oracle Cloud Object Storage (PAR Asset Sync), PostgreSQL (pgvector/pg_trgm), Linux Systems Governance, Nginx, SSL/TLS

Core Technical Projects

Master's Industry Experience Project (Monash University)

Subang Jaya, Malaysia

ENTERPRISE THREAT INTELLIGENCE & INTELLIGENT PROCESSING ENGINE

2026

- **Retrieval-Augmented Classification (RAC) Pipeline:** Architected a dynamic fusion pipeline orchestrating domain-specific transformers and calibrated XGBoost heads. Applied INT8 dynamic quantization strictly to transformer weights to isolate and resolve CPU compute bottlenecks, while preserving FP32 precision on the 70k HNSW index to maintain absolute threat-detection recall. Achieved stable end-to-end processing latencies between **0.095s** and **0.181s**.
- **Explainable AI (XAI) & Matrix Ablation:** Engineered a zero-latency XAI layer utilizing Leave-One-Out (LOO) matrix ablation and native C++ SHAP feature attribution. Developed a custom Anisotropy Cone correction using dynamic mean-centering for unpooled transformer tokens, enabling high-variance semantic heatmaps and mathematically grounded semantic validation.
- **ARM-Optimized Computer Vision Engine:** Built an adaptive OCR pipeline using RapidOCR (PP-OCRv5) on ARM64, integrating width-bucketed batching, dynamic quality routing, and geometric array filtering. Retained FP32 precision for the vision models after benchmarking revealed INT8 quantization introduced operator conversion overhead that regressed inference speed, maintaining an end-to-end system latency (inclusive of RAC analysis) of **0.6s** to **1.8s**.
- **Systems Engineering & Production Hardening:** Deployed a stateless production engine on Oracle A1 using ONNX Runtime with hardcoded `intra_op_num_threads` and custom Docker `shm_size` to completely eliminate Linux CFS context-switching lag. Engineered automated, lean deployments by dynamically syncing heavy ML models from Oracle Object Storage via Pre-Authenticated Requests (PAR) into isolated named volumes.

Professional Experience

Myrmidon Laboratories

Singapore

ROBOTICS ENGINEER

Apr 2022 - Sep 2023

- **Acoustic Data Transmission:** Developed audible-spectrum data modulation protocols over solid barriers to enable RF-free, secure localized communication paths.
- **Algorithmic Spatial Localization:** Built a deterministic tracking pipeline for low-resolution infrared sensor arrays (8×8 to 64×64 pixels) to localize occupancy without predictive models.
- **Interactive Systems Engineering:** Programmed a lightweight ESP32 scheduling engine using native C++, utilizing MQTT payloads to trigger synchronized hardware execution patterns.

Azure Innovations

Kuala Lumpur, Malaysia

MACHINE LEARNING ENGINEER

Sep 2020 - Feb 2022

- **Local Identity Verification:** Developed an end-to-end vision pipeline pairing YOLOv4 object detection with deep metric learning for facial feature extraction.
- **Vector Space Matching:** Engineered a local matching system to compute Cosine distances between live detection frames and registered user embeddings.
- **Standalone Pipeline Execution:** Configured the deep learning workloads to run entirely within local machine environments, eliminating cloud API dependencies.